## 1. Introduction

Air pollution forecasting plays a crucial role in environmental monitoring, public health planning, and urban sustainability. Accurately predicting pollutant concentrations allow governments and communities to anticipate dangerous conditions, implement preventive measures, and minimize population exposure to harmful emissions.

In this project, we predict the concentrations of key atmospheric pollutants using the Air Quality UCI dataset, which contains 9357 hourly observations of air quality recorded in an Italian city during 2004-2005. The model focuses on five major pollutants, CO(GT), NMHC(GT), NOx (GT), and NO2(GT), and predicts their value 1, 6, 12, and 24 hours into the future.

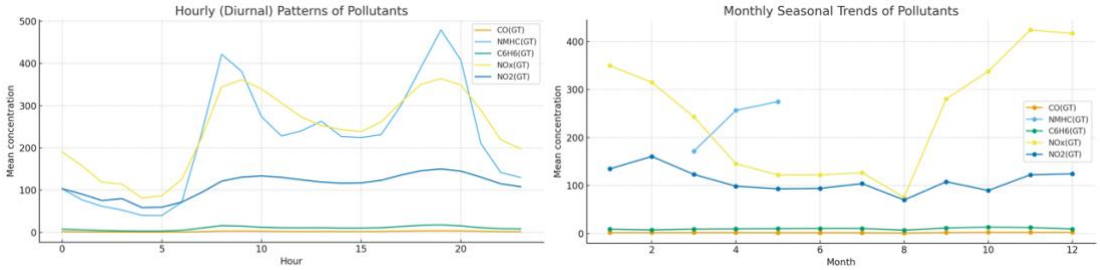
The goal of this project is to evaluate the effectiveness of at least 2 regression and 2 classification models in forecasting multiple pollutants simultaneously at several time horizons, analyze its performance relative to baseline methods, and understand the challenges associated with multivariate air quality prediction.

## 2. Data Analysis

In terms of data cleaning, we first replaced all sentinel −200 values with NaN for all sensor and pollutant variables. We then quantified missingness per column and observed that NMHC(GT) has over 90% missing values, while CO(GT), NOx(GT) and NO2(GT) have roughly 15–20% missing, and the remaining sensors and meteorological variables have only small gaps. Because the dataset is hourly and relatively smooth, we applied time-based linear interpolation over short gaps and then forward-/backward-filled any remaining leading or trailing NaNs, instead of dropping large parts of the series. After imputation, we merged Date and Time into a single timestamp, sorted the dataset chronologically, and generated time features such as hour of day, weekday and month to capture daily and seasonal cycles. All five pollutants are presented in diurnal patterns. The periods of 8-10AM and 5-8 PM correspond to morning and evening traffic rush hours. Concentration decreases during the night, reaching minimum levels between 12-7 AM. This pattern shows that vehicular emissions are a dominant factor of short-term pollutant variation.

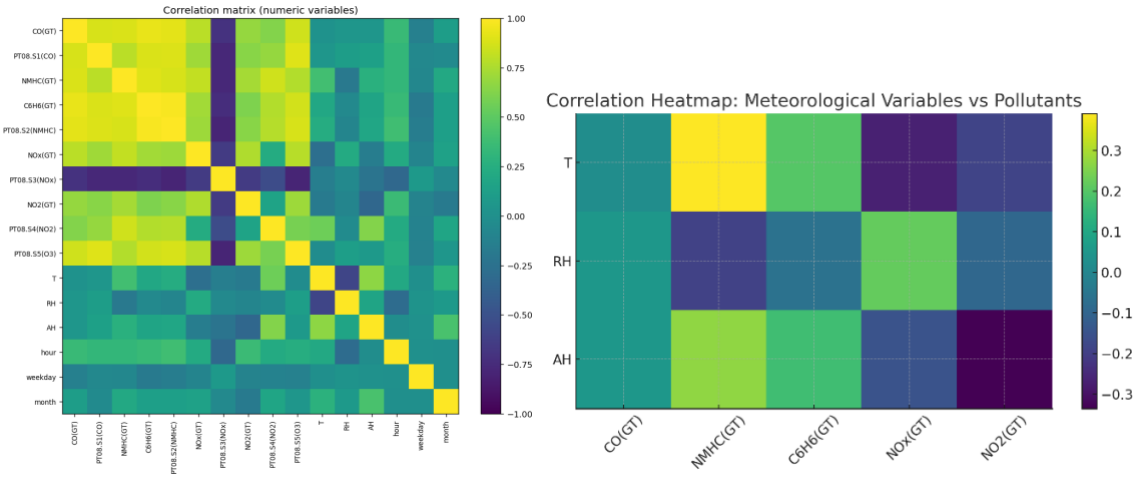
Seasonally, pollutants show a big difference between winter and summer. NO2 and NOx are substantially higher during colder months, whereas concentrations decline in summer.

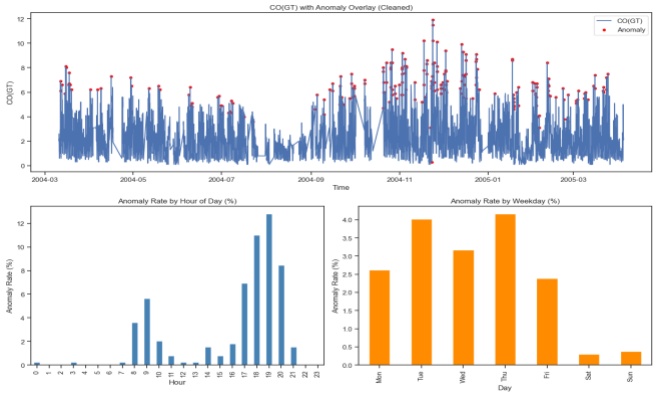
CO and C6H6 show similar trends, there is no big fluctuation throughout the year. NMHC only appears between March and May because most of the data are missing.




In the numeric variables heatmap, temperature and absolute humidity are strongly positively correlated, but relative humidity is negatively correlated with T and AH. Hour correlates moderately with pollutants, reflecting rush hour patterns. Weekday show small correlations, showing lower pollution levels on weekends. Month correlates with both pollutants and meteorology, capturing seasonal behaviour. The strong dark purple tiles correspond to variables with fixed missing value patterns.

The correlation heatmap between meteorological variables and pollutants shows how meteorological conditions influence pollutant concentrations. NMHC and C6H6 show positive correlations while NOx and NO2 show negative correlation with temperature, which means that warmer weather enhances the release of NMHC and C6H6. NOx and NO2 are accumulated in cold and stable air, especially during winter. NOx dispersion is limited in cold, damp weather, but NMHC and C6H6 tend to evaporate less during higher humidity. Absolute humidity strongly positively correlates with temperature and therefore shares some temperature relationships.






## 3. Methodology

We used the AirQualityUCI dataset containing sensor readings and meteorological features recorded hourly. We removed empty unnamed columns created by extra delimiters, handled missing values using the interpolation and fill strategy described in Section 2, converted Date and Time into a single timestamp, and sorted the dataset chronologically. In terms of feature engineering, to capture temporal dependence, we created lagged versions of key pollutant variables with 1, 6, 12, 24 hours. These features help models learn short-term and long-term trends. For each pollutant, CO values were distributed into 3 classes for classification. We used all data up to the end of 2024 as training, all 2025 data for testing, and the last 10 percent of the training period for validation. We also used StandardScaler and fitted it only on the training data, then applied the same transformation to validation and test sets.

We selected Linear Regression, XGBoost, and LSTM to represent three levels of model complexity. Linear Regression serves as a simple baseline, we choose it because it is a classic statistical model for continuous prediction, fast to train and easy to interpret. XGBoost is one of the most powerful models for tabular time-series forecasting. It can capture nonlinear relationships and feature interactions, handle high-dimensional engineered features, also be robust to noise and missing values. LSTM is designed to model sequential temporal dependencies, specifically built for time-series forecasting, and can also learn long-term dependencies across multiple hours.

In terms of classification models, we chose Logistic Regression as a baseline classifier and compared it to Random Forest and XGBoost. Logistic Regression provides a simple benchmark for classifying CO into three levels, it is fast, simple, widely used and works reasonably well when the decision boundary is close to linear. Random Forest is suitable

for irregular or nonlinear class boundaries. XGBoost is one of the strongest models for classification with numerical features, it can capture complex nonlinear patterns and has very strong predictive accuracy on tabular classification tasks.

## 4. Results

We evaluated three regression models, namely Linear Regression, XGBoost, and LSTM, across five pollutants and four prediction horizons. Performance was measured using Root Mean Squared Error.

Overall, XGBoost achieved the lowest RMSE across most pollutants and horizons, proving its effectiveness for tabular time-series prediction. However, LSTM underperformed, generating higher RMSE than both XGBoost and Linear Regression for nearly all pollutants.

| Models | Pollutants | t+1 | t+6 | t+12 | t+24 |
|---|---|---|---|---|---|
| Linear Regression | CO | 0.6543 | 1.2514 | 1.1004 | 1.0977 |
| | NO2 | 24.0782 | 53.7943 | 45.0511 | 46.1896 |
| | NOx | 99.5553 | 218.2561 | 203.2089 | 191.3195 |
| | C6H6 | 3.2039 | 5.5833 | 5.1108 | 5.2828 |
| | NMHC | 9.6239 | 19.2884 | 18.4324 | 29.9196 |
| XGBoost | CO | 0.6146 | 1.0272 | 1.2217 | 1.1872 |
| | NO2 | 24.1088 | 40.5652 | 38.9030 | 41.2358 |
| | NOx | 100.7726 | 205.0583 | 177.1999 | 179.5573 |
| | C6H6 | 3.0515 | 5.4137 | 6.5313 | 6.8014 |
| | NMHC | 13.7679 | 42.2421 | 8.5080 | 15.0755 |
| LSTM | CO | 1.1293 | 1.3363 | 1.3631 | 1.4654 |
| | NO2 | 40.5162 | 47.4964 | 51.8589 | 52.0733 |
| | NOx | 171.3288 | 210.4055 | 203.8210 | 215.4250 |
| | C6H6 | 5.5986 | 7.3225 | 6.4766 | 7.8667 |
| | NMHC | 75.3101 | 74.2737 | 66.3230 | 113.6122 |

We evaluated three classification models, namely Logistic Regression, Random Forest, and XGBoost. To predict CO concentration levels at forecasting horizons of 1, 6, 12, 24 hours. Performance was assessed using accuracy, as well as precision, recall, and F1-score to account for class imbalance across the three classes.

Logistic Regression provided reasonable performance at t+1 but degraded significantly for longer horizons. Random Forest consistently outperformed the baseline. It achieved moderate improvement in precision, recall, and F1-score. XGBoost achieved the best

overall performance across all forecasting horizons. It shows that gradient boosting effectively captures complex relationships.

| Models | Classification report | t+1 | t+6 | t+12 | t+24 |
|---|---|---|---|---|---|
| Logistic Regression | Accuracy | 0.65 | 0.39 | 0.41 | 0.38 |
| | Macro avg (precision) | 0.65 | 0.33 | 0.45 | 0.47 |
| | Macro avg (recall) | 0.64 | 0.38 | 0.42 | 0.38 |
| | Macro avg (F1-score) | 0.63 | 0.27 | 0.41 | 0.33 |
| XGBoost | Accuracy | 0.73 | 0.56 | 0.53 | 0.57 |
| | Macro avg (precision) | 0.71 | 0.55 | 0.51 | 0.57 |
| | Macro avg (recall) | 0.71 | 0.54 | 0.50 | 0.57 |
| | Macro avg (F1-score) | 0.71 | 0.54 | 0.48 | 0.56 |
| Random Forest | Accuracy | 0.71 | 0.55 | 0.50 | 0.53 |
| | Macro avg (precision) | 0.69 | 0.52 | 0.47 | 0.51 |
| | Macro avg (recall) | 0.68 | 0.52 | 0.47 | 0.50 |
| | Macro avg (F1-score) | 0.68 | 0.49 | 0.44 | 0.47 |

## 5. Discussion

XGBoost outperformed all other models in both regression and classification tasks and reasons for strong performance is that it handles nonlinear interactions between lagged values, and is robust to noise and outliers, also captures threshold behavior useful for CO classification, leading to the lowest RMSE for almost all pollutants and the highest classification accuracy across all horizons.

We also examined model residuals to identify potential anomalies and short-term pollution events. For our best CO regression model, we flagged hours where the absolute prediction error exceeded roughly three times the training RMSE. These large residuals usually coincided with sharp spikes in CO and NOx during morning and evening rush hours or very cold nights. This suggests that many of the “anomalies” correspond to genuine high-pollution episodes rather than pure sensor faults. In a real deployment, such residual-based checks could be used to raise alerts for unusual behaviour and to support more robust monitoring.

Sensor drift may suffer from sensitivity degradation, temperature or humidity interference, leading to NMHC’s noise and very high RMSE across models and inconsistent behavior of NOx across horizons.

For classification, CO levels were not evenly distributed among three classes. As a result, Logistic Regression struggled heavily at higher horizons. In real deployment, oversampling can be applied to solve data imbalance issues.

In terms of temporal concept drift, air pollution shows seasonal and behavioral variation, the emissions are higher on weekdays than weekends and are lower in winter.

## 6. Conclusion

This project investigated the problem of short and long-term air quality forecasting using the AirQualityUCI dataset. Three regression models and three classification models were evaluated across four prediction horizons. The results show that XGBoost consistently provided the strongest performance in both regression and classification tasks.

Linear Regression served as a baseline, performing well for pollutants with stable short-term time such as CO and C6H6, but degrading as horizon increased. LSTM, despite its theoretical advantage for sequential modelling, underperformed across all pollutants because of the small dataset, high levels of sensor noise, and limited temporal structure in the pollutants. Random Forest provided moderate classification performance but was outperformed by XGBoost. The study also points out several important limitations that affect real-world deployment. Sensor drift and measurement noise, especially in NMHC and NOx, significantly reduced predictive accuracy across all models. Class imbalance affected classification performance for CO events, requiring class-balancing strategies for solving imbalanced problems.

Overall, the findings suggest that tree-based ensemble methods, particularly XGBoost, are the most suitable modelling approach for this type of dataset. Future improvements could include incorporating additional meteorological variables, applying sensor drift compensation, and adding multi-year datasets. These improvements would help ensure reliable and robust air-quality forecasting in real operational environments.

## 7. References

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